Reflective Journal

Overview

Working through this data preparation lab made me realize that preprocessing isn't just a cleanup step, it's where we shape raw information into a format models can actually understand. Raw data is messy and full of missing details. Preparing it properly ensures our machine learning models perform fairly and accurately.

What I Learned & Key Insights

1. Imputing Missing Data

I used to think deleting rows with missing data was the easiest solution. However, doing that throws away valuable information. When handling missing values in the Age column, using the median made much more sense than the mean. The median isn't pulled off-course by extreme values (like very old or very young passengers), making it a safer pick for skewed data.

2. Converting Text to Numbers (One-Hot Encoding)

Models operate on numbers, not words. To convert categorical text like Sex or Embarked into numbers, I used One-Hot Encoding. I learned that simply assigning numbers (like 1, 2, 3) creates a fake order or ranking that tricks the model. Creating binary columns (1 or 0) solves this. Using drop_first=True also prevents redundant data by dropping extra columns the model can easily infer.

3. Scaling Numerical Features

Features like Age (0–80) and Fare (0–512) live on completely different scales. Without scaling, a model might assume higher numbers in Fare are naturally more important. Using StandardScaler puts both features on equal footing centered around a mean of zero. I also

realized that distance-based models need scaling, but tree-based models don't because they split data using simple "greater than" comparisons on individual features.

Problem Solving & Technical Lessons

While coding Task 1, I encountered a Pandas FutureWarning because I used inplace=True. I learned that Pandas is deprecating this syntax because chained indexing can create unpredictable copies in memory. I fixed this by using explicit reassignment instead: df['Age'] = df['Age'].fillna(median_age). This taught me the importance of writing future-proof code.

Final Thoughts

My biggest takeaway is that data preparation requires thoughtful decisions at every turn. Choosing how to fill missing values or encode variables directly impacts how well a model learns. Data preparation isn't just administrative work; it forms the core foundation of machine learning.